# Climate, Health & Mortality

## Research Context and the APHRC Approach

**Flavian Otieno** | March 17, 2026

African Population and Health Research Center
Transforming lives in Africa through research.

# Outline

**01**

### Why Climate & Health?

*The public health case for understanding climate-driven mortality*

**02**

### The Climate–Mortality Pathway

*How temperature and rainfall become health risks*

**03**

### APHRC & the INSPIRE Network

*Who we are and how we generate evidence across Africa*

**04**

### Our Data

*A brief about climate and health dataset*

**05**

### The Challenge Opportunity

*Why machine learning can help*

# Why Does This Research Matter?

**250,000**

additional deaths per year projected from climate change by 2030

**90%**

of climate-related deaths occur in low- and middle-income countries

**Sub-Saharan Africa**

most vulnerable region - least equipped with evidence to respond

*Climate change is not just an environmental issue — it is a health emergency.*

# How Climate Becomes a Health Risk

### Rising Temperatures

Heat stress, dehydration, worsening chronic disease

→

### Shifting Rainfall

Flooding, drought, disrupted food & water systems

→

### Ecological Change

Expanding vector habitats, new disease geographies

↓ **Health Outcomes: Increased Mortality · Infectious Disease · Malnutrition · Mental Health Burden**

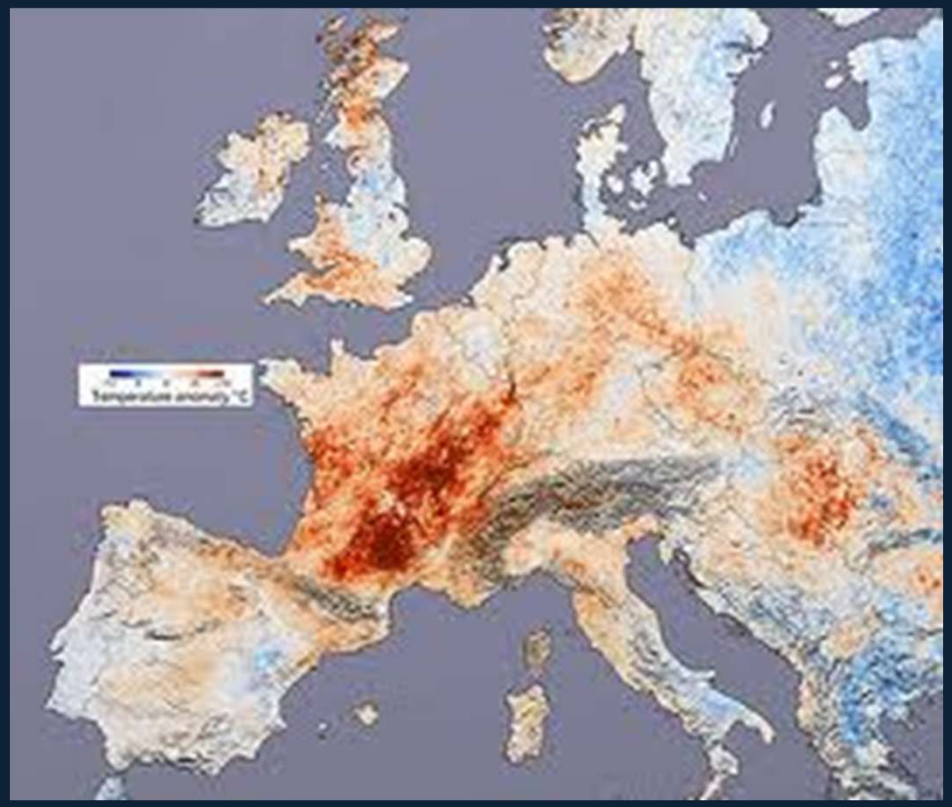
# Linking Climate and Health Data

- Limited integration of longitudinal Health Demographic and Surveillance System (HDSS) data with climate datasets to assess health impacts of climate change.
- Despite HDSS data being used for policymaking, its potential for analyzing climate-driven health outcomes remains underutilized.
- Linking longitudinal HDSS data with climate data is essential for:

✓ Understanding how climate change impacts population health over time

✓ Developing evidence-based strategic to mitigate climate-related health risks

✓ Improving planning and resource allocation to address climate-sensitive health issues



# APHRC & the INSPIRE Network

## APHRC

*African Population and Health Research Center*

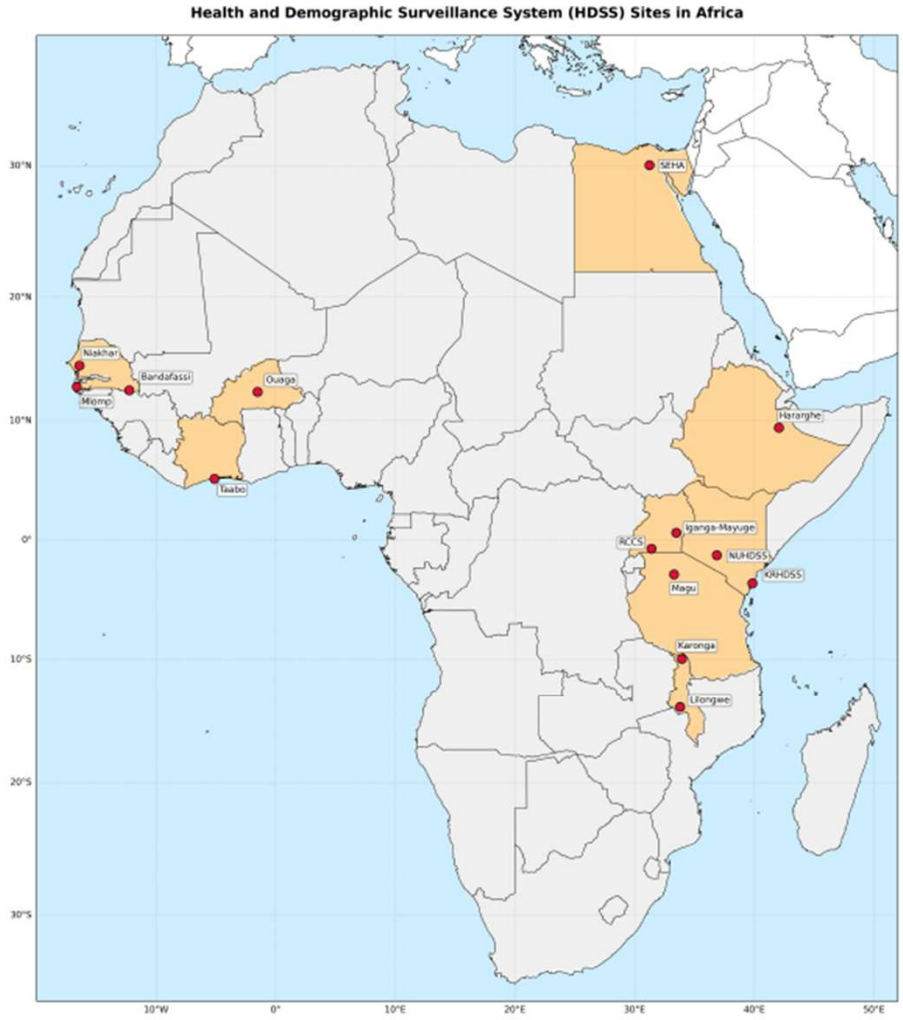
- Pan-African research institution based in Nairobi, Kenya
- Vision – Transform lives in Africa through research
- Mission – Generating evidence, strengthening research and related capacity in the African research and development ecosystem, and engaging policy to inform action on health and development
- Committed to open and FAIR data principles

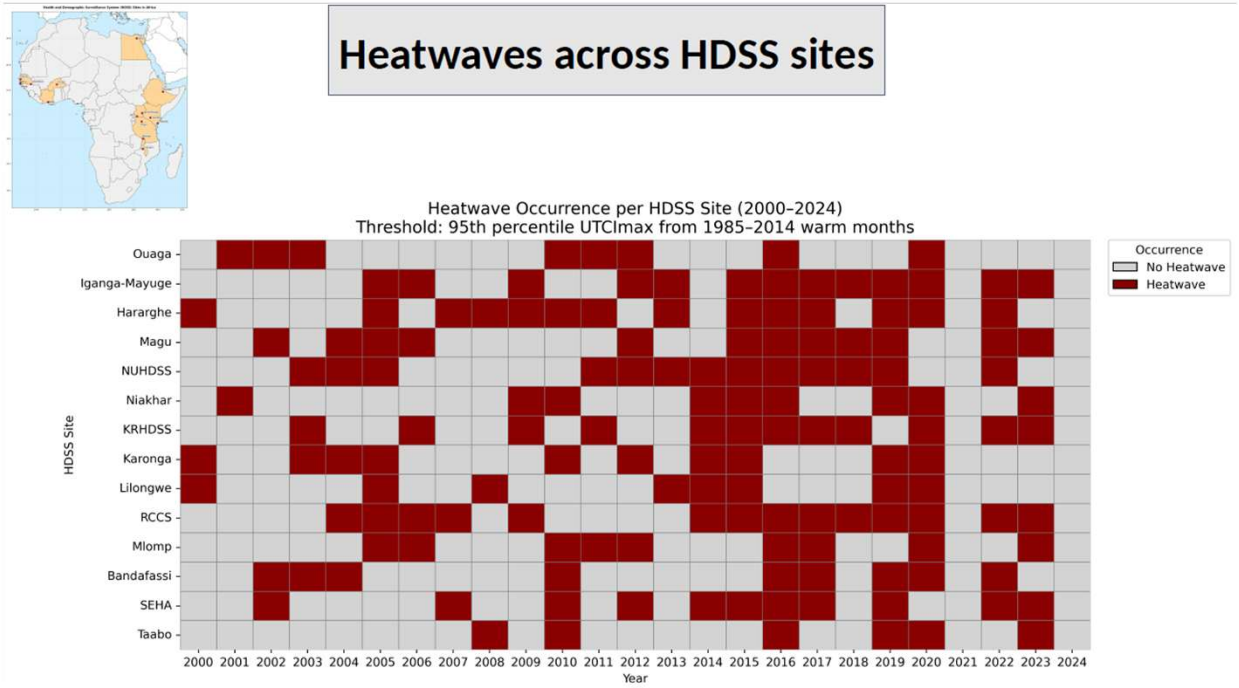
## INSPIRE Network

*Implementation Network for Sharing Population Information from Research Entities (INSPIRE)*

- Network of HDSS sites across Africa
- Longitudinal population data spanning decades
- Tracks births, deaths, migration, morbidity, pregnancies, household characteristics
- Unique asset: individual-level linkage across time
- Enables climate exposure to be linked to health events

# HDSS Under INSPIRE



# How Data is collected

## Health & Mortality Data

Health and Demographic Surveillance System (HDSS) sites collects individual life events — births, deaths, cause of death, migration — continuously and longitudinally across participating communities.

## Climate Data

Daily temperature, rainfall, relative humidity and other climate variables are sourced from ground stations and remote-sensing climate data

## Linkage & Harmonisation

Climate exposures are linked to individual health records, creating longitudinal climate-health datasets ready for analysis.

*Result: A uniquely rich, longitudinal, population-level dataset linking climate to mortality across African settings.*

# Why Machine Learning?

## The Challenge

- Non-linear relationships between climate & health
- Lagged effects — exposure today, death weeks later
- Spatial heterogeneity across diverse geographies
- Interaction effects between multiple climate stressors

## What ML Can Do

- Detect complex, non-linear exposure-response patterns
- Identify mortality signals across climate and time
- Handle high-dimensional data with many covariates
- Generalise patterns across sites with different contexts
- Deliver insights that translate to policy action

# Thank You

## Questions & Discussion

**Flavian Otieno** | Data Scientist

# About the Challenge

AI4EAC INNOVATION CHALLENGE

# Challenge Design

**Climate-Sensitive Mortality Prediction Challenge**

**The task:** Build a machine learning model to identify which deaths are linked to climate-sensitive causes.

**Data:**

- Tabular
- Additional weather data required!

**Multi-Metric Evaluation:**

- **F1-Score (50%):** Balances precision and recall for imbalance data
- **ROC-AUC (50%):** Measures ranking quality across all decision thresholds.
- Requires both binary predictions and probability

**Submission Format requirements:**

- Binary target (0/1) using default threshold of 0.5
- Probability scores (0.0 to 1.0)
- Custom thresholds are **not permitted**

EAST AFRICAN SCIENCE & TECHNOLOGY COMMISSION
EASTECO

AI4EAC INNOVATION CHALLENGE

# Preparing for the Challenge

**Practice Challenge:** [The AI4EAC Health Practice Challenge]

**The task:** Predict who is suffering from depression based on routine survey data

**Evaluation Metric:**

- **Mean Absolute Error:** Measures the percentage of survey respondents incorrectly predicted as “depressed” (the closer to zero the better)
- Requires only binary predictions

**Key Preparation activities:**

- Familiarise yourself with Zindi
- Understand the submission process
- Handling class imbalance

EAST AFRICAN SCIENCE & TECHNOLOGY COMMISSION
EASTECO

AI4EAC INNOVATION CHALLENGE

AI4EAC
is here!
AI4
EAC
INNOVATION
CHALLENGE